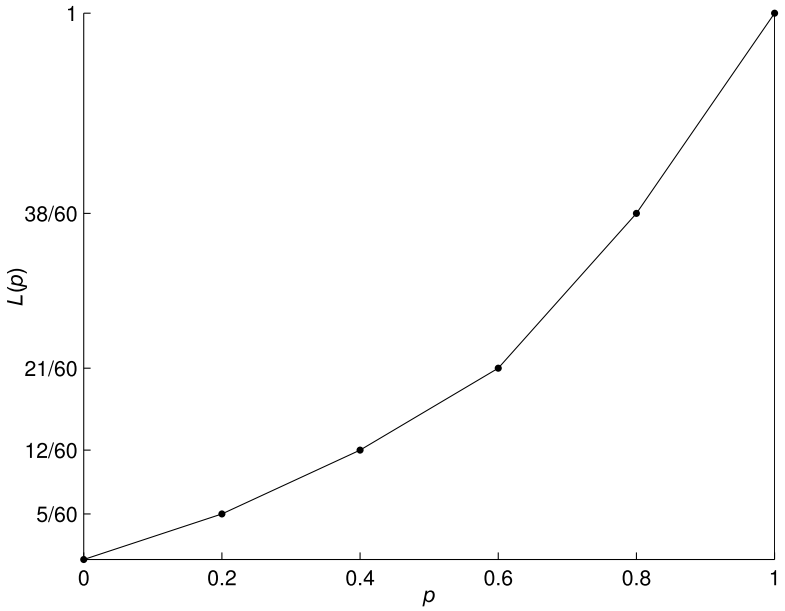


FIGURE 2.16
The Lorenz curve for Example 2.7.a.

$$L(.4) = 12/60,$$
$$L(.6) = 21/60,$$
$$L(.8) = 38/60,$$
$$L(1) = 60/60.$$

Connecting with straight lines between these points gives the Lorenz curve (Figure 2.16). ■

Lorenz curves are also used to plot the wealths of members of a population, with $L(p)$ now representing the proportion of all wealth owned by the least wealthy $100p$ percent of the population. For instance, if the population consists of 1000 people, then $L(.22)$ is the proportion of the populations's wealth that is owned by its 220 poorest members.

Now, the average of the numbers in a set always increases when a new value that is larger than any of the previous ones is added to the set. Consequently, if the incomes, in increasing value, of the members of the population are $x_1 \leq x_2 \leq x_3 \cdots \leq x_n$, then for any $j = 1, \ldots, n$

$$\frac{x_1 + \ldots + x_j}{j} \leq \frac{x_1 + \ldots + x_{j+1}}{j+1} \leq \ldots \leq \frac{x_1 + \ldots + x_n}{n},$$

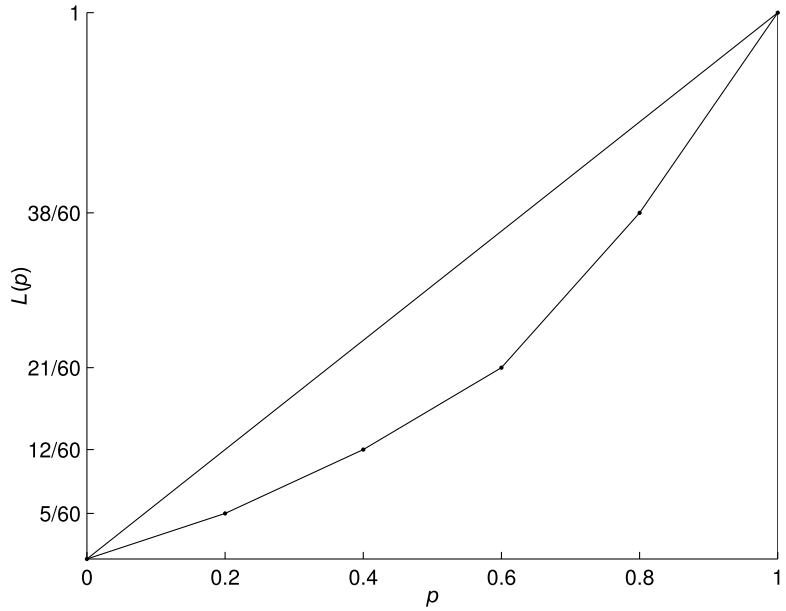


FIGURE 2.17
The Lorenz curve and the straight line.

implying that

$$\frac{x_1 + \ldots + x_j}{x_1 + \ldots + x_n} \leq \frac{j}{n}.$$

Thus,

$$L(j/n) \leq j/n.$$

In addition, it can be verified that $L(j/n) = j/n$, $j = 1, \ldots, n$, only when all n incomes are equal. Consequently, unless all incomes are equal, the Lorenz curve is always below the straight line from $(0, 0)$ to $(1, 1)$. (Figure 2.17 indicates this for the data of Example 2.7.a.)

Thus, the Lorenz curve is equal to the straight line from $(0, 0)$ to $(1, 1)$ when all incomes are equal and dips below the line when incomes are not all equal. The more unequal the incomes the greater the "hump", equal to the region between the straight line and the Lorenz curve (the shaded region in Figure 2.18).

A measure of the hump and thus of the inequality of the incomes is given by the *Gini index*, which is the ratio of the area of the hump divided by the area under the straight line. Because the area of a triangle is one half its base times its height, the area under the straight line from $(0, 0)$ to $(1, 1)$ is equal to $1/2$.

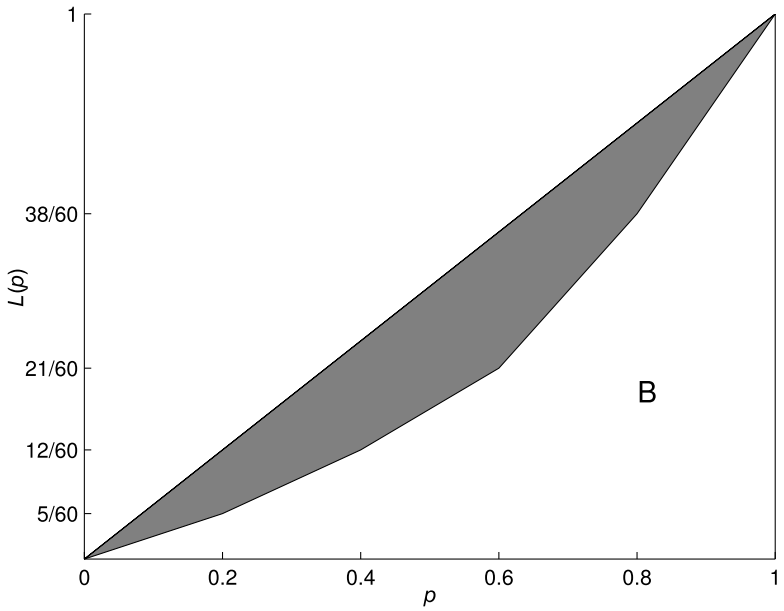


FIGURE 2.18
The hump of the Lorenz curve.

Hence, the Gini index, call it G, is given by

$$G = \frac{\text{area of the hump of the Lorenz curve}}{1/2}.$$

Letting B be the area beneath the Lorenz curve, and using that the area of the hump is equal to the area under the straight line minus the area under the Lorenz curve, we see that the value of the Gini index is

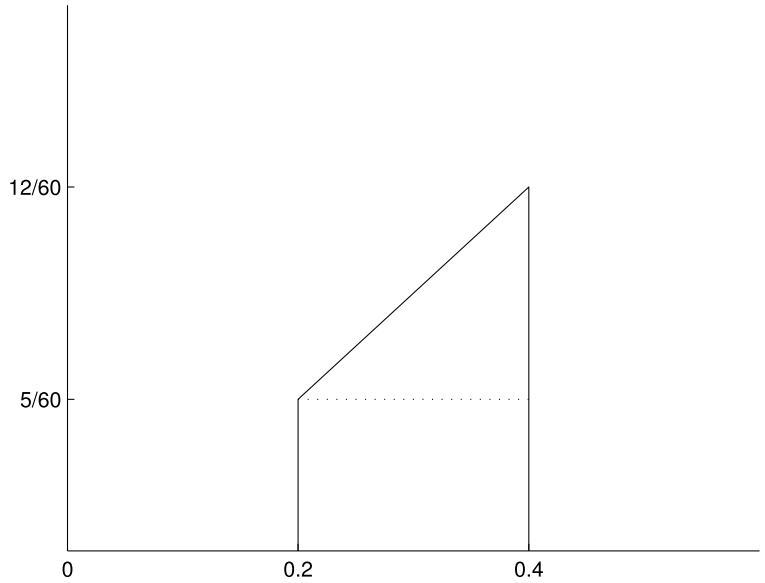
$$G = \frac{1/2 - B}{1/2} = 1 - 2B.$$

Example 2.7.b. Compute the Gini index for the data of Example 2.7.a.

Solution. To compute B, the area under the Lorenz curve, we write $B = B_1 + B_2 + B_3 + B_4 + B_5$, where B_1 is the area of the region under the Lorenz curve from 0 to .2; B_2 is the area of the region between .2 and .4; B_3 is the area of the region between .4 and .6; B_4 is the area of the region between .6 and .8; and B_5 is the area of the region between .8 and 1. Now, B_1 is the area of a triangle whose base is .2 and whose height is 5/60, showing that

$$B_1 = (1/2)(.2)(5/60) = 5/600.$$

Now, B_2, B_3, B_4, and B_5 are the areas of regions composed of a triangle on top of a rectangle. For instance, the following is the region B_2:



Now all of the triangles in these regions have base .2 and respective heights $5/60, 7/60, 9/60, 17/60, 22/60$. Consequently, the sum of the areas of all these triangles is

$$\text{Area of triangles} = \frac{1}{2}(.2)(5+7+9+17+22)/60 = .1.$$

The four rectangles all have base length .2 and respective heights 5/60, 12/60, 21/60, 38/60, yielding

$$\text{Area of rectangles} = .2(5+12+21+38)/60 \approx .25333.$$

Adding the area of the rectangles and the area of the triangles gives $B = .1 + .25333 = .35333$, and thus

$$G = 1 - .70666 = .29334.$$

In general, suppose the incomes are, in increasing order, $x_1 \le x_2 \le x_3 \cdots \le x_n$. Let

$$s_j = x_1 + \ldots + x_j, \quad j = 1, \ldots, n.$$

Then all the triangles will have base $1/n$ and will have respective heights $x_1/s_n, x_2/s_n, \ldots, x_n/s_n$. Consequently, the sum of the area of all the triangles is

$$\text{Area of triangles} = \frac{1}{2n}(x_1 + \ldots + x_n)/s_n = \frac{1}{2n}.$$

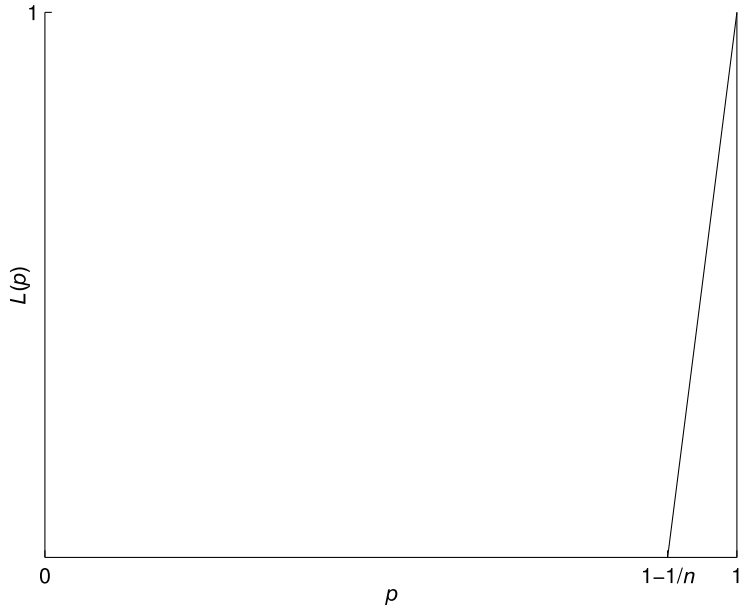


FIGURE 2.19
Largest possible Gini index value.

The rectangles will all have length $1/n$ and will have respective heights $s_1/s_n, s_2/s_n, \ldots, s_{n-1}/s_n$. Hence,

$$\text{Area of rectangles} = \frac{s_1 + s_2 + \ldots + s_{n-1}}{ns_n},$$

giving

$$B = \frac{1}{2n} + \frac{s_1 + s_2 + \ldots + s_{n-1}}{ns_n}. \quad \blacksquare$$

The Gini index is 0 when all incomes are equal (and so the area between the straight line and the Lorenz curve is 0). On the opposite extreme, the largest possible value of the Gini index occurs when only one of the n members of the population has a positive income. In this case, the area under the Lorenz curve is the area of a triangle whose base is of length $1/n$ and whose height is 1 (see Figure 2.19). Thus, $B = \frac{1}{2n}$, and $G = 1 - 1/n$.

2.8 Using R

Suppose you want the sample mean and sample variance of the values $x_1, \ldots, x_n$. Type
$x = c(x_1, \ldots, x_n)$

and hit return. [Interpret $x = c(x_1, \ldots, x_n)$ as the sequence of values $x_1, \ldots, x_n$.]
On the next line type
mean(x)
and when you hit return, the sample mean is displayed. On the next line, type
var(x)
and when you hit return, the variance is displayed.

For instance, suppose you wanted the sample mean and sample variance of the values 4, 6, 12, 9, 21, 14. Then

```
> x=c(4,6,12,9,21,14)
> mean(x)
[1] 11
> var(x)
[1] 37.6
```

By the way, you do not type > at the beginning of a line. R always starts a line with this symbol, which is a prompt asking what to do next. (The [1] in front of the answers returned is to indicate that you only asked for a single item.)

Other commands are sum(x), which returns the sum of the values; median(x), which returns the sample median; and sd(x), which returns the sample standard deviation (equal to the square root of var(x)). Also, if $x = c(x_1, \ldots, x_n)$ and $y = c(y_1, \ldots, y_n)$, then $x + y = c(x_1 + y_1, \ldots, x_n + y_n)$, $x - y = c(x_1 - y_1, \ldots, x_n - y_n)$, $x * y = c(x_1 y_1, \ldots, x_n y_n)$, and if all $y_i \neq 0$, $x/y = c(x_1/y_1, \ldots, x_n/y_n)$. Also, $x^2 = x * x = c(x_1^2, \ldots, x_n^2)$. If a is a real number, then $a + x = c(a + x_1, \ldots, a + x_n)$, $a * x = c(ax_1, \ldots, ax_n)$ and, for $a \neq 0$, $x/a = c(x_1/a, \ldots, x_n/a)$. In addition, if all $x_i \neq 0$, then $a/x = c(a/x_1, \ldots, a/x_n)$.

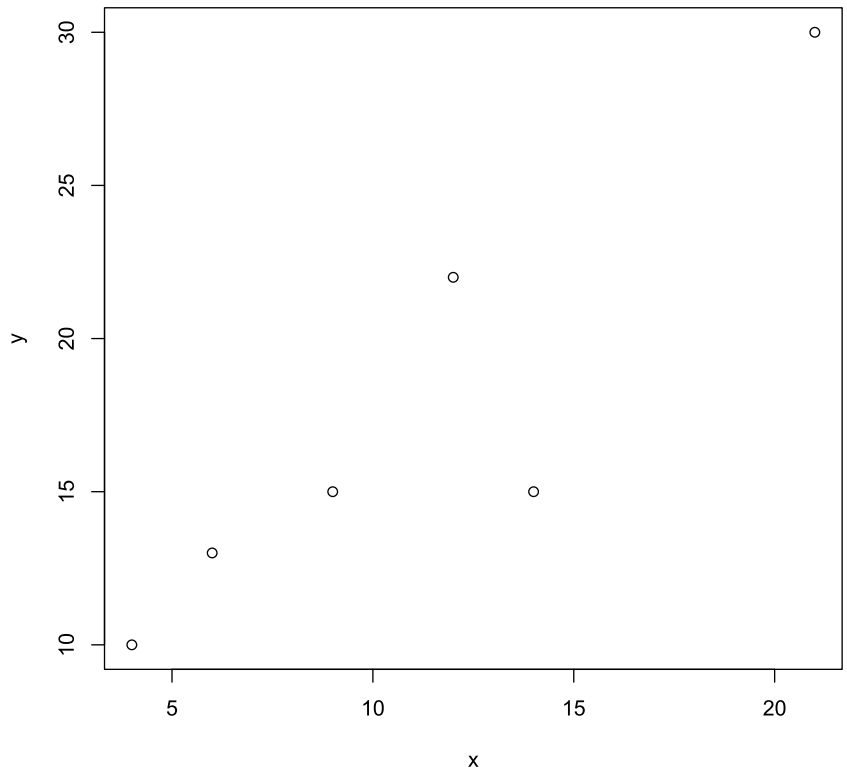
If you have data pairs $(x_1, y_1), \ldots, (x_n, y_n)$, then to obtain the sample correlation coefficient, do the following:

```
> x = c(x_1, ..., x_n)
> y = c(y_1, ..., y_n)
> cor(x, y)
```

To obtain a scatter diagram of the data, type
> plot(x,y)
and hit return.

For instance, suppose we have the data pairs (4, 10), (6, 13), (12, 22), (9, 15), (21, 30), (14, 15), then we obtain the sample correlation coefficient and scatter diagram as follows:

```
> x = c(4, 6, 12, 9, 21, 14)
> y = c(10, 13, 22, 15, 30, 15)
> cor(x, y)
[1] 0.9041494
> plot(x,y)
```



R can also be used for computation. For instance, suppose we wanted to compute $\frac{18\sqrt{177}}{677}$. This is accomplished by

$>$ 18 $*$ sqrt(177)/677
[1] 0.3537288

Suppose now that we want to select an element of the vector $x = c(x_1, \ldots, x_n)$, say x_i, its value in position i. This is accomplished by the R command $> x[i]$. If we wanted the new vector, consisting of the elements of the vector x, going from positions i to j, use the R command $x[i : j]$.

For instance,

$> x = c(3, 18, 9, 7, 22, 5, 17)$
$> x[3]$
[1] 9
$> x[2 : 5]$
[1]18 9 7 22